

AGA CoStays

Complete Final Requirement, API, Entity & UI Flow Document

Multi-Branch Hotel + Restaurant Management System

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This is the final complete document created from scratch for AGA CoStays. It includes system requirements, final role hierarchy, access rules, customer public browsing flow, login-protected actions, multi-city and multi-branch hotel logic, location-based room pricing, room/hotel/dish photos, restaurant workflow, payments, billing, feedback, attendance, payroll, reports, analytics, notifications, database entities, API lists, request samples, and UI screens in a clear flow.

1. System Overview

- Application Name: AGA CoStays.
- System Type: Multi-branch hotel chain platform available across metropolitan cities.
- Main Purpose: Customers can browse hotels by city/location, view hotel and room photos, book rooms, order restaurant food from the room, pay bills, and submit feedback.
- Frontend: React responsive web application.
- Backend: REST API-based services/microservices.
- Database: MySQL or PostgreSQL relational database.
- Authentication: JWT Bearer token.
- Authorization: Role-Based Access Control with branch-level restrictions.
- Payment Gateway: Razorpay/Stripe/PayPal-style gateway integration.
- Email Service: SMTP/SendGrid/AWS SES-style service for booking confirmation, payment, checkout reminder, feedback, and salary notifications.

2. Final Role Hierarchy

```
ROOT_ADMIN
├── MANAGER
│   ├── RECEPTIONIST
│   ├── RESTAURANT_ADMIN
│   ├── CHEF
│   ├── SERVING_STAFF
│   └── HOUSEKEEPING_STAFF
└── CUSTOMER
```

3. Role Responsibilities

Role	Responsibility
ROOT_ADMIN	Full website/platform owner. Can manage managers, roles, global settings, payment/email/SMS settings, all branches, audit logs, backups, exports, analytics and every business detail.
MANAGER	Business operations admin under Root Admin. Replaces old ADMIN role. Manages cities, branches, staff, rooms, room prices, bookings, restaurant, payroll, attendance and reports.
CUSTOMER	Can browse the website without login. Must login/register only for booking, food order,

	payment, support request, feedback, profile and own data.
RECEPTIONIST	Branch-level staff. Can manage assigned branch rooms, photos/descriptions/status, bookings, check-in/checkout, food order assistance, attendance and branch analytics.
RESTAURANT_ADMIN	Branch restaurant-level admin. Can manage assigned branch restaurant menu, dish photos, food orders, chef/serving staff assignments and restaurant reports.
CHEF	Assigned branch kitchen staff. Can view assigned kitchen orders, start preparation, mark food ready, add notes and mark attendance.
SERVING_STAFF	Assigned branch delivery staff. Can view delivery orders, pick up food, mark delivered, report issues and mark attendance.
HOUSEKEEPING_STAFF	Assigned branch housekeeping staff. Can view assigned room cleaning tasks, update cleaning status and mark attendance.

4. Customer Public Browsing and Login Rule

- Customer can view the website without logging in.
- Customer can browse Home, Cities, Hotel Branches, Hotel Photos, Room List, Room Details, Room Photos, Location-Based Room Prices, Restaurant Menu, Dish Photos, Receptionist Contacts and Emergency Contacts without login.
- When Customer clicks Book Room, Order Food, Make Payment, View My Bookings, Submit Feedback, Raise Support Request, View Profile or View Notifications, the system must show Login/Register page.
- After successful login or registration, redirect the customer back to the previous selected page/action.
- Example: Customer selects Bengaluru > Deluxe Room > Book Now. If not logged in, redirect to login/register. After login, return to the same booking page.

4.1 Public Customer APIs

```

GET /api/website/banners
GET /api/cities
GET /api/hotel-branches
GET /api/hotel-branches/city/{cityId}
GET /api/hotel-branches/{branchId}
GET /api/hotel-branches/{branchId}/photos
GET /api/hotel-branches/{branchId}/rooms
GET /api/hotel-branches/{branchId}/rooms/{roomId}
GET /api/hotel-branches/{branchId}/rooms/{roomId}/photos
GET /api/hotel-branches/{branchId}/rooms/availability
GET /api/hotel-branches/{branchId}/restaurant
GET /api/hotel-branches/{branchId}/restaurant/menu
GET /api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}
GET /api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}/photos
GET /api/hotel-branches/{branchId}/receptionists
GET /api/hotel-branches/{branchId}/emergency-contacts
GET /api/website/about-us
GET /api/website/contact-us
GET /api/website/privacy-policy
GET /api/website/terms-and-conditions

```

4.2 Login Required Customer APIs

```

POST /api/hotel-branches/{branchId}/bookings
GET /api/bookings/my-bookings
GET /api/hotel-branches/{branchId}/bookings/{bookingId}
PUT /api/bookings/{bookingId}/cancel
POST /api/hotel-branches/{branchId}/restaurant/orders
GET /api/restaurant/orders/my-orders
GET /api/restaurant/orders/{orderId}
PUT /api/restaurant/orders/{orderId}/cancel
POST /api/payments/create
POST /api/payments/verify
GET /api/payments/my-payments
GET /api/billing/my-bill/{bookingId}
POST /api/billing/final-bill/{bookingId}/pay
GET /api/billing/final-bill/{bookingId}/invoice
POST /api/customer-support/requests
GET /api/customer-support/requests/my-requests
POST /api/feedback

```

```
POST /api/restaurant/feedback
GET /api/customers/profile
PUT /api/customers/profile
GET /api/notifications/my-notifications
```

5. Important Business Rules

- Root Admin is the highest platform owner.
- Manager is under Root Admin and replaces the old Admin role.
- Customers self-register; Manager should not manually create customer accounts.
- Manager creates staff accounts and assigns them to branches.
- Each hotel branch has its own rooms, photos, receptionist contacts, restaurant, dishes, dish photos, orders, attendance, payroll and reports.
- Receptionist, Restaurant Admin, Chef, Serving Staff and Housekeeping Staff can access only their assigned branch.
- Customers can view all public branch/location data but can access only their own bookings, orders, bills, payments and feedback.
- Booking confirmation email should be sent after booking and must include branch, room, payment, receptionist and emergency contact details.
- Checkout reminder email should be sent 3 hours before checkout.
- After checkout, feedback email should be sent for hotel stay and restaurant food taste.
- Payroll/salary APIs are only for Manager and Root Admin.
- Root Admin can access everything including root settings, audit logs and backups.

6. Location-Based Room Pricing

- Default base prices exist: NORMAL = 800/day, DELUXE = 1600/day, LUXURY = 2500/day.
- Manager can update current room price based on branch/location, demand, season or business rule.
- Root Admin can update price for any branch.
- Receptionist cannot update price by default.
- Customers always see currentPricePerDay for the selected branch.
- Booking must use currentPricePerDay and store it as price_per_day_at_booking.
- Existing confirmed bookings should not be affected by future price changes.
- Every price change must be stored in room_price_history.

Branch	NORMAL	DELUXE	LUXURY
AGA CoStays Bengaluru	1000/day	1800/day	3000/day
AGA CoStays Chennai	800/day	1600/day	2500/day
AGA CoStays Mumbai	1500/day	2500/day	4000/day

7. Constants / Enums

Role Values:

ROOT_ADMIN, MANAGER, CUSTOMER, RECEPTIONIST, RESTAURANT_ADMIN, CHEF, SERVING_STAFF, HOUSEKEEPING_STAFF

RoomStatus:

AVAILABLE, BOOKED, OCCUPIED, CLEANING, MAINTENANCE

BookingStatus:

PENDING, APPROVED, REJECTED, CHECKED_IN, CHECKED_OUT, CANCELLED

FoodOrderStatus:

PLACED, ACCEPTED, PREPARING, READY_FOR_DELIVERY, OUT_FOR_DELIVERY, DELIVERED, CANCELLED, REJECTED

PaymentStatus:

PENDING, SUCCESS, FAILED, REFUNDED, PAY_AT_CHECKOUT

AttendanceStatus:

PRESENT, ABSENT, HALF_DAY, ON_LEAVE, WEEK_OFF, HOLIDAY

PayrollStatus:

GENERATED, PENDING, PROCESSING, PAID, FAILED, ON_HOLD, CANCELLED

PaymentMode:

BANK_TRANSFER, UPI, CASH, CHEQUE

FoodType:

VEG, NON_VEG, EGG, VEGAN

MenuCategory:

STARTER, MAIN_COURSE, RICE, BIRYANI, BREADS, DESSERT, BEVERAGE, SNACKS, BREAKFAST, LUNCH, DINNER

Department:

HOTEL_FRONT_DESK, RESTAURANT_KITCHEN, RESTAURANT_SERVICE, HOUSEKEEPING, ADMINISTRATION

Shift:

MORNING, EVENING, NIGHT

8. Authentication APIs

Method	Endpoint	Access	Purpose
POST	/api/auth/register	Public / CUSTOMER	Customer self-registration
POST	/api/auth/login	All roles	Login and receive JWT token
POST	/api/auth/logout	All authenticated roles	Logout current session
POST	/api/auth/forgot-password	Public	Send password reset link
POST	/api/auth/reset-password	Public	Reset password using token

9. ROOT_ADMIN APIs

Method	Endpoint	Access	Purpose
POST	/api/root-admin/auth/login	ROOT_ADMIN	Root Admin login
POST	/api/root-admin/auth/logout	ROOT_ADMIN	Root Admin logout
PUT	/api/root-admin/change-password	ROOT_ADMIN	Change Root Admin password
POST	/api/root-admin/managers	ROOT_ADMIN	Create Manager
GET	/api/root-admin/managers	ROOT_ADMIN	Get all Managers
GET	/api/root-admin/managers/{managerId}	ROOT_ADMIN	Get Manager by ID
PUT	/api/root-admin/managers/{managerId}	ROOT_ADMIN	Update Manager
PUT	/api/root-admin/managers/{managerId}/status	ROOT_ADMIN	Activate/deactivate Manager
DELETE	/api/root-admin/managers/{managerId}	ROOT_ADMIN	Delete Manager
POST	/api/root-admin/roles	ROOT_ADMIN	Create role
GET	/api/root-admin/roles	ROOT_ADMIN	Get roles
PUT	/api/root-admin/roles/{roleId}	ROOT_ADMIN	Update role
DELETE	/api/root-admin/roles/{roleId}	ROOT_ADMIN	Delete role
POST	/api/root-admin/roles/{roleId}/permissions	ROOT_ADMIN	Assign permissions
GET	/api/root-admin/roles/{roleId}/permissions	ROOT_ADMIN	Get permissions
GET	/api/root-admin/website/settings	ROOT_ADMIN	Get website settings
PUT	/api/root-admin/website/settings	ROOT_ADMIN	Update website settings
PUT	/api/root-admin/website/booking-status	ROOT_ADMIN	Enable/disable booking
PUT	/api/root-admin/website/restaurant-ordering-status	ROOT_ADMIN	Enable/disable restaurant ordering
PUT	/api/root-admin/website/maintenance-mode	ROOT_ADMIN	Enable/disable maintenance mode
POST	/api/root-admin/website/banners	ROOT_ADMIN	Add website banner
GET	/api/website/banners	Public, ROOT_ADMIN	Get website banners
PUT	/api/root-admin/website/banners/{bannerId}	ROOT_ADMIN	Update banner
DELETE	/api/root-admin/website/banners/{bannerId}	ROOT_ADMIN	Delete banner
PUT	/api/root-admin/website/about-us	ROOT_ADMIN	Update About Us
PUT	/api/root-admin/website/terms-and-conditions	ROOT_ADMIN	Update Terms
PUT	/api/root-admin/website/privacy-policy	ROOT_ADMIN	Update Privacy Policy
GET	/api/root-admin/hotel-branches	ROOT_ADMIN	Get all branches
GET	/api/root-admin/hotel-branches/{branchId}/complete-details	ROOT_ADMIN	Get full branch details
PUT	/api/root-admin/hotel-branches/{branchId}/status	ROOT_ADMIN	Activate/deactivate branch
GET	/api/root-admin/dashboard	ROOT_ADMIN	Root dashboard
GET	/api/root-admin/analytics/website	ROOT_ADMIN	Website analytics
GET	/api/root-admin/analytics/branches/performance	ROOT_ADMIN	Branch performance

GET	/api/root-admin/analytics/cities/performance	ROOT_ADMIN	City performance
GET	/api/root-admin/analytics/customers/growth	ROOT_ADMIN	Customer growth
GET	/api/root-admin/analytics/revenue	ROOT_ADMIN	Platform revenue analytics
GET	/api/root-admin/settings/payment-gateway	ROOT_ADMIN	Get payment settings
PUT	/api/root-admin/settings/payment-gateway	ROOT_ADMIN	Update payment settings
GET	/api/root-admin/settings/email	ROOT_ADMIN	Get email settings
PUT	/api/root-admin/settings/email	ROOT_ADMIN	Update email settings
GET	/api/root-admin/settings/sms	ROOT_ADMIN	Get SMS settings
PUT	/api/root-admin/settings/sms	ROOT_ADMIN	Update SMS settings
GET	/api/root-admin/audit-logs	ROOT_ADMIN	Get audit logs
GET	/api/root-admin/audit-logs/{auditLogId}	ROOT_ADMIN	Get audit log
GET	/api/root-admin/export/customers	ROOT_ADMIN	Export customers
GET	/api/root-admin/export/bookings	ROOT_ADMIN	Export bookings
GET	/api/root-admin/export/revenue	ROOT_ADMIN	Export revenue
GET	/api/root-admin/export/payroll	ROOT_ADMIN	Export payroll
POST	/api/root-admin/backups	ROOT_ADMIN	Create backup
GET	/api/root-admin/backups	ROOT_ADMIN	Get backups
GET	/api/root-admin/backups/{backupId}/download	ROOT_ADMIN	Download backup

10. MANAGER APIs

Method	Endpoint	Access	Purpose
POST	/api/manager/roles	MANAGER	Create staff role
GET	/api/manager/roles	MANAGER	Get roles
GET	/api/manager/roles/{roleId}	MANAGER	Get role by ID
PUT	/api/manager/roles/{roleId}	MANAGER	Update role
DELETE	/api/manager/roles/{roleId}	MANAGER	Delete role
POST	/api/manager/staff	MANAGER	Create staff account
GET	/api/manager/staff	MANAGER	Get staff
GET	/api/manager/staff/{staffId}	MANAGER	Get staff by ID
PUT	/api/manager/staff/{staffId}	MANAGER	Update staff
PUT	/api/manager/staff/{staffId}/assign-role	MANAGER	Assign role
PUT	/api/manager/staff/{staffId}/status	MANAGER	Activate/deactivate staff
DELETE	/api/manager/staff/{staffId}	MANAGER	Delete staff
GET	/api/manager/customers	MANAGER	Get customers
GET	/api/manager/customers/{customerId}	MANAGER	Get customer by ID
PUT	/api/manager/customers/{customerId}/status	MANAGER	Activate/deactivate customer
POST	/api/manager/cities	MANAGER	Create city
PUT	/api/manager/cities/{cityId}	MANAGER	Update city
DELETE	/api/manager/cities/{cityId}	MANAGER	Deactivate city
POST	/api/manager/hotel-branches	MANAGER	Create hotel branch
PUT	/api/manager/hotel-branches/{branchId}	MANAGER	Update branch
DELETE	/api/manager/hotel-branches/{branchId}	MANAGER	Deactivate branch
POST	/api/manager/hotel-branches/{branchId}/photos	MANAGER	Upload branch photos
PUT	/api/manager/hotel-branches/{branchId}/photos/{photoId}	MANAGER	Update branch photo
PUT	/api/manager/hotel-branches/{branchId}/photos/{photoId}/replace	MANAGER	Replace branch photo
PUT	/api/manager/hotel-branches/{branchId}/photos/{photoId}/primary	MANAGER	Set primary branch photo
DELETE	/api/manager/hotel-branches/{branchId}/photos/{photoId}	MANAGER	Delete branch photo
POST	/api/manager/hotel-branches/{branchId}/staff	MANAGER	Assign staff to branch
PUT	/api/manager/staff/{staffId}/transfer-branch	MANAGER	Transfer staff branch
DELETE	/api/manager/hotel-branches/{branchId}/staff/{staffId}	MANAGER	Remove staff from branch
PUT	/api/manager/hotel-branches/{branchId}/rooms/{roomId}/price	MANAGER, ROOT_ADMIN	Update room price
PUT	/api/manager/hotel-branches/{branchId}/room-types/{roomType}/price	MANAGER, ROOT_ADMIN	Bulk update room type price
GET	/api/manager/hotel-branches/{branchId}/rooms/{roomId}/price-history	MANAGER, ROOT_ADMIN	View price history
GET	/api/manager/analytics/branches/dashboard	MANAGER	All branches dashboard
GET	/api/manager/analytics/branches/revenue	MANAGER	Branch revenue
GET	/api/manager/analytics/branches/occupancy	MANAGER	Branch occupancy
GET	/api/manager/analytics/branches/attendance	MANAGER	Branch attendance

11. City, Hotel Branch and Staff Branch APIs

Method	Endpoint	Access	Purpose
GET	/api/cities	Public, CUSTOMER, MANAGER, ROOT_ADMIN	Get active cities
GET	/api/hotel-branches	Public, CUSTOMER, MANAGER, ROOT_ADMIN	Get hotel branches
GET	/api/hotel-branches/city/{cityId}	Public, CUSTOMER, MANAGER, ROOT_ADMIN	Get branches by city

GET	/api/hotel-branches/{branchId}	Public, CUSTOMER, Assigned Staff, MANAGER, ROOT_ADMIN	Get branch details
GET	/api/hotel-branches/{branchId}/photos	Public, CUSTOMER, Assigned Staff, MANAGER, ROOT_ADMIN	Get branch photos
GET	/api/hotel-branches/{branchId}/receptionists	CUSTOMER, Assigned Staff, MANAGER, ROOT_ADMIN	Get receptionist contacts
GET	/api/hotel-branches/{branchId}/emergency-contacts	CUSTOMER, Assigned Staff, MANAGER, ROOT_ADMIN	Get emergency contacts
GET	/api/staff/my-branch	Branch Staff	Get logged-in staff branch

12. Branch Room, Room Photo and Pricing APIs

Method	Endpoint	Access	Purpose
POST	/api/hotel-branches/{branchId}/rooms	Assigned RECEPTIONIST, MANAGER	Add room
GET	/api/hotel-branches/{branchId}/rooms	Public, CUSTOMER, Assigned Staff, MANAGER, ROOT_ADMIN	Get branch rooms with prices/photos
GET	/api/hotel-branches/{branchId}/rooms/{roomId}	Public, CUSTOMER, Assigned Staff, MANAGER, ROOT_ADMIN	Get room details
GET	/api/hotel-branches/{branchId}/rooms/availability	Public, CUSTOMER, Assigned Staff, MANAGER, ROOT_ADMIN	Check availability
PUT	/api/hotel-branches/{branchId}/rooms/{roomId}	Assigned RECEPTIONIST, MANAGER	Update room excluding price
PUT	/api/hotel-branches/{branchId}/rooms/{roomId}/description	Assigned RECEPTIONIST, MANAGER	Update room description
PUT	/api/hotel-branches/{branchId}/rooms/{roomId}/status	Assigned RECEPTIONIST, HOUSEKEEPING_STAFF, MANAGER	Update room status
POST	/api/hotel-branches/{branchId}/rooms/{roomId}/photos	Assigned RECEPTIONIST, MANAGER	Upload room photos
GET	/api/hotel-branches/{branchId}/rooms/{roomId}/photos	Public, CUSTOMER, Assigned Staff, MANAGER, ROOT_ADMIN	View room photos
PUT	/api/hotel-branches/{branchId}/rooms/{roomId}/photos/{photoId}	Assigned RECEPTIONIST, MANAGER	Update room photo
PUT	/api/hotel-branches/{branchId}/rooms/{roomId}/photos/{photoId}/replace	Assigned RECEPTIONIST, MANAGER	Replace room photo
PUT	/api/hotel-branches/{branchId}/rooms/{roomId}/photos/{photoId}/primary	Assigned RECEPTIONIST, MANAGER	Set primary room photo
DELETE	/api/hotel-branches/{branchId}/rooms/{roomId}/photos/{photoId}	Assigned RECEPTIONIST, MANAGER	Delete room photo

13. Branch Booking APIs

Method	Endpoint	Access	Purpose
POST	/api/hotel-branches/{branchId}/bookings	CUSTOMER	Create booking in selected branch
POST	/api/hotel-branches/{branchId}/bookings/customer/{customerId}	Assigned RECEPTIONIST, MANAGER	Create booking for customer
GET	/api/bookings/my-bookings	CUSTOMER	Get own bookings
GET	/api/hotel-branches/{branchId}/bookings	Assigned RECEPTIONIST, MANAGER	Get branch bookings
GET	/api/hotel-branches/{branchId}/bookings/{bookingId}	CUSTOMER own, Assigned RECEPTIONIST, MANAGER	Get booking details
PUT	/api/hotel-branches/{branchId}/bookings/{bookingId}/approve	Assigned RECEPTIONIST, MANAGER	Approve booking
PUT	/api/hotel-branches/{branchId}/bookings/{bookingId}/reject	Assigned RECEPTIONIST, MANAGER	Reject booking
PUT	/api/bookings/{bookingId}/cancel	CUSTOMER own, Assigned RECEPTIONIST, MANAGER	Cancel booking
PUT	/api/hotel-branches/{branchId}/bookings/{bookingId}/check-in	Assigned RECEPTIONIST, MANAGER	Check-in
PUT	/api/hotel-branches/{branchId}/bookings/{bookingId}/check-out	Assigned RECEPTIONIST, MANAGER	Checkout

14. Restaurant, Dish Photo and Food Order APIs

Method	Endpoint	Access	Purpose
GET	/api/hotel-branches/{branchId}/restaurant	Public, CUSTOMER, Assigned Staff, MANAGER	Get restaurant details
POST	/api/hotel-branches/{branchId}/restaurant/menu	Assigned RESTAURANT_ADMIN, MANAGER	Add dish
GET	/api/hotel-branches/{branchId}/restaurant/menu	Public, CUSTOMER, Assigned Staff, MANAGER	Get menu with dish photos
GET	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}	Public, CUSTOMER, Assigned Staff, MANAGER	Get dish details
PUT	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}	Assigned RESTAURANT_ADMIN, MANAGER	Update dish
DELETE	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}	Assigned RESTAURANT_ADMIN, MANAGER	Delete/soft-delete dish
PUT	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}/availability	Assigned RESTAURANT_ADMIN, MANAGER	Update dish availability
POST	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}/photos	Assigned RESTAURANT_ADMIN, MANAGER	Upload dish photos
GET	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}/photos	Public, CUSTOMER, Assigned Staff, MANAGER	View dish photos
PUT	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}/photos/{photoId}	Assigned RESTAURANT_ADMIN, MANAGER	Update dish photo
PUT	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}/photos/{photoId}/replace	Assigned RESTAURANT_ADMIN, MANAGER	Replace dish photo
PUT	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}/photos/{photoId}/primary	Assigned RESTAURANT_ADMIN, MANAGER	Set primary dish photo

	branches/{branchId}/restaurant/menu/{menuItemId}/photos/{photoId}/primary	MANAGER	
DELETE	/api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}/photos/{photoId}	Assigned RESTAURANT_ADMIN, MANAGER	Delete dish photo
POST	/api/hotel-branches/{branchId}/restaurant/orders	CUSTOMER	Place food order
POST	/api/hotel-branches/{branchId}/restaurant/orders/customer/{customerId}	Assigned RECEPTIONIST, MANAGER	Place order for customer
GET	/api/restaurant/orders/my-orders	CUSTOMER	Get own food orders
GET	/api/hotel-branches/{branchId}/restaurant/orders	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Get branch orders
GET	/api/restaurant/orders/{orderId}	CUSTOMER own, Assigned Staff, MANAGER	Get order details
PUT	/api/hotel-branches/{branchId}/restaurant/orders/{orderId}/accept	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Accept order
PUT	/api/hotel-branches/{branchId}/restaurant/orders/{orderId}/reject	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Reject order
PUT	/api/restaurant/orders/{orderId}/cancel	CUSTOMER own, Assigned Staff, MANAGER	Cancel order
PUT	/api/hotel-branches/{branchId}/restaurant/orders/{orderId}/assign-chef	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Assign same-branch chef
PUT	/api/hotel-branches/{branchId}/restaurant/orders/{orderId}/assign-serving-staff	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Assign same-branch serving staff

15. Chef, Serving Staff and Housekeeping APIs

Method	Endpoint	Access	Purpose
GET	/api/staff/my-branch/restaurant/kitchen/orders	CHEF	Get kitchen orders
PUT	/api/staff/my-branch/restaurant/kitchen/orders/{orderId}/start-preparing	CHEF	Start preparing food
PUT	/api/staff/my-branch/restaurant/kitchen/orders/{orderId}/mark-ready	CHEF	Mark food ready
PUT	/api/staff/my-branch/restaurant/kitchen/orders/{orderId}/notes	CHEF	Add kitchen notes
GET	/api/staff/my-branch/restaurant/serving/orders/my-orders	SERVING_STAFF	Get delivery orders
PUT	/api/staff/my-branch/restaurant/serving/orders/{orderId}/pickup	SERVING_STAFF	Pick up food
PUT	/api/staff/my-branch/restaurant/serving/orders/{orderId}/delivered	SERVING_STAFF	Mark delivered
POST	/api/staff/my-branch/restaurant/serving/orders/{orderId}/issue	SERVING_STAFF	Report delivery issue
GET	/api/staff/my-branch/housekeeping/tasks/my-tasks	HOUSEKEEPING_STAFF	Get cleaning tasks
PUT	/api/staff/my-branch/housekeeping/tasks/{taskId}/status	HOUSEKEEPING_STAFF	Update cleaning status
POST	/api/hotel-branches/{branchId}/housekeeping/tasks	Assigned RECEPTIONIST, MANAGER	Create cleaning task
GET	/api/hotel-branches/{branchId}/housekeeping/tasks	Assigned RECEPTIONIST, MANAGER	Get cleaning tasks

16. Attendance APIs

Method	Endpoint	Access	Purpose
POST	/api/staff/my-branch/attendance/check-in	Branch Staff	Attendance check-in
POST	/api/staff/my-branch/attendance/check-out	Branch Staff	Attendance check-out
POST	/api/attendance/mark	MANAGER, Assigned RECEPTIONIST	Manual attendance
POST	/api/attendance/mark-absent	MANAGER, Assigned RECEPTIONIST	Mark absent
GET	/api/hotel-branches/{branchId}/attendance/today/working	MANAGER, Assigned RECEPTIONIST	Working staff today
GET	/api/hotel-branches/{branchId}/attendance/today/absent	MANAGER, Assigned RECEPTIONIST	Absent staff today
GET	/api/hotel-branches/{branchId}/attendance/monthly	MANAGER, Assigned RECEPTIONIST	Monthly branch attendance
GET	/api/attendance/staff/{staffId}	Own Staff, MANAGER, Assigned RECEPTIONIST	Staff attendance
GET	/api/attendance/staff/{staffId}/monthly	Own Staff, MANAGER, Assigned RECEPTIONIST	Staff monthly attendance
PUT	/api/attendance/{attendanceId}	MANAGER, Assigned RECEPTIONIST	Update attendance
DELETE	/api/attendance/{attendanceId}	MANAGER	Delete attendance

17. Payment APIs

Method	Endpoint	Access	Purpose
POST	/api/payments/create	CUSTOMER, RECEPTIONIST, MANAGER	Create payment
POST	/api/payments/verify	CUSTOMER, RECEPTIONIST, MANAGER	Verify payment
GET	/api/payments/my-payments	CUSTOMER	My payments
GET	/api/payments/{paymentId}	CUSTOMER own, RECEPTIONIST, MANAGER	Payment by ID
GET	/api/payments/booking/{bookingId}	CUSTOMER own, RECEPTIONIST, MANAGER	Payments by booking
POST	/api/payments/{paymentId}/refund	MANAGER	Refund payment
POST	/api/payments/webhook	Payment Gateway/System	Payment webhook
POST	/api/restaurant/payments/create	CUSTOMER, RECEPTIONIST, MANAGER	Create food payment
POST	/api/restaurant/payments/verify	CUSTOMER, RECEPTIONIST, MANAGER	Verify food payment
GET	/api/restaurant/payments/order/{orderId}	CUSTOMER own, RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Food payment by order
POST	/api/restaurant/orders/{orderId}/add-to-	CUSTOMER, RECEPTIONIST, MANAGER	Add food to final bill

	final-bill		
--	------------	--	--

18. Billing and Feedback APIs

Method	Endpoint	Access	Purpose
GET	/api/billing/final-bill/{bookingId}	RECEPTIONIST, MANAGER	Generate final bill
GET	/api/billing/my-bill/{bookingId}	CUSTOMER own	View own bill
POST	/api/billing/final-bill/{bookingId}/pay	CUSTOMER, RECEPTIONIST, MANAGER	Pay final bill
GET	/api/billing/final-bill/{bookingId}/invoice	CUSTOMER own, RECEPTIONIST, MANAGER	Download invoice
POST	/api/feedback	CUSTOMER	Submit hotel/restaurant feedback
GET	/api/feedback/my-feedback	CUSTOMER	My feedback
GET	/api/feedback	MANAGER	All feedback
GET	/api/feedback/booking/{bookingId}	RECEPTIONIST, MANAGER	Feedback by booking
POST	/api/restaurant/feedback	CUSTOMER	Submit restaurant feedback
GET	/api/restaurant/feedback/my-feedback	CUSTOMER	My restaurant feedback
GET	/api/restaurant/feedback	RESTAURANT_ADMIN, RECEPTIONIST, MANAGER	Restaurant feedback
GET	/api/restaurant/feedback/order/{orderId}	RESTAURANT_ADMIN, RECEPTIONIST, MANAGER	Feedback by order

19. Payroll / Salary APIs

Only Manager and Root Admin can access payroll APIs. Other staff and customers have no salary/payroll access.

Method	Endpoint	Access	Purpose
POST	/api/manager/payroll/salary-structures	MANAGER	Create salary structure
GET	/api/manager/payroll/salary-structures	MANAGER	Get salary structures
GET	/api/manager/payroll/salary-structures/role/{roleName}	MANAGER	Get salary by role
PUT	/api/manager/payroll/salary-structures/{salaryStructureId}	MANAGER	Update salary structure
DELETE	/api/manager/payroll/salary-structures/{salaryStructureId}	MANAGER	Deactivate salary structure
POST	/api/manager/payroll/staff/{staffId}/bank-account	MANAGER	Add staff bank account
GET	/api/manager/payroll/staff/{staffId}/bank-account	MANAGER	Get bank account
PUT	/api/manager/payroll/staff/{staffId}/bank-account	MANAGER	Update bank account
DELETE	/api/manager/payroll/staff/{staffId}/bank-account	MANAGER	Deactivate bank account
POST	/api/manager/payroll/generate	MANAGER	Generate payroll
POST	/api/manager/payroll/staff/{staffId}/generate	MANAGER	Generate staff payroll
GET	/api/manager/payroll	MANAGER	Get monthly payroll
GET	/api/manager/payroll/{payrollId}	MANAGER	Get payroll by ID
GET	/api/manager/payroll/staff/{staffId}	MANAGER	Payroll by staff
PUT	/api/manager/payroll/{payrollId}	MANAGER	Update payroll before payment
POST	/api/manager/payroll/credit-salaries	MANAGER	Credit all salaries
POST	/api/manager/payroll/staff/{staffId}/credit-salary	MANAGER	Credit one salary
GET	/api/manager/payroll/{payrollId}/payment-status	MANAGER	Salary payment status
POST	/api/manager/payroll/{payrollId}/retry-payment	MANAGER	Retry salary payment
POST	/api/manager/payroll/{payrollId}/salary-slip	MANAGER	Generate salary slip
GET	/api/manager/payroll/{payrollId}/salary-slip	MANAGER	Download salary slip
POST	/api/manager/payroll/{payrollId}/send-salary-slip-email	MANAGER, SYSTEM	Send salary slip email
GET	/api/manager/payroll/reports/monthly-summary	MANAGER	Payroll monthly summary
GET	/api/manager/payroll/reports/by-role	MANAGER	Payroll by role
GET	/api/manager/payroll/reports/by-department	MANAGER	Payroll by department
GET	/api/manager/payroll/reports/payments	MANAGER	Salary payment report
POST	/api/manager/hotel-branches/{branchId}/payroll/generate	MANAGER	Generate branch payroll
POST	/api/manager/hotel-branches/{branchId}/payroll/credit-salaries	MANAGER	Credit branch salaries
GET	/api/manager/hotel-branches/{branchId}/payroll	MANAGER	Get branch payroll
GET	/api/manager/payroll/reports/branch-wise-summary	MANAGER	Branch payroll summary

20. Notification / Email APIs

Method	Endpoint	Access	Purpose
POST	/api/notifications/booking-confirmation-email	SYSTEM, RECEPTIONIST, MANAGER	Send booking confirmation
GET	/api/notifications/booking/{bookingId}/email-preview	CUSTOMER own, RECEPTIONIST, MANAGER	Preview booking email
POST	/api/notifications/booking/{bookingId}/resend-email	CUSTOMER own, RECEPTIONIST, MANAGER	Resend booking email

POST	/api/notifications/checkout-reminder/schedule	SYSTEM	Schedule checkout reminder
POST	/api/notifications/checkout-reminder/send	SYSTEM	Send checkout reminder
GET	/api/notifications/checkout-reminder/booking/{bookingId}	CUSTOMER own	Get checkout reminder
POST	/api/notifications/feedback-email	SYSTEM, RECEPTIONIST, MANAGER	Send feedback email
POST	/api/notifications/payment-confirmation	SYSTEM	Send payment email
POST	/api/restaurant/notifications/order-confirmation-email	SYSTEM	Send food order email
POST	/api/restaurant/notifications/food-delivered	SYSTEM	Food delivered notification
POST	/api/manager/payroll/notifications/salary-credited	SYSTEM, MANAGER	Salary credited notification
GET	/api/notifications/my-notifications	CUSTOMER	Customer notifications
PUT	/api/notifications/{notificationId}/read	CUSTOMER	Mark notification read

21. Reports and Analytics APIs

Method	Endpoint	Access	Purpose
GET	/api/hotel-branches/{branchId}/analytics/hotel/dashboard	Assigned RECEPTIONIST, MANAGER	Branch hotel dashboard
GET	/api/hotel-branches/{branchId}/analytics/hotel/occupancy	Assigned RECEPTIONIST, MANAGER	Branch occupancy
GET	/api/hotel-branches/{branchId}/analytics/hotel/bookings	Assigned RECEPTIONIST, MANAGER	Branch bookings analytics
GET	/api/hotel-branches/{branchId}/analytics/hotel/room-type-revenue	Assigned RECEPTIONIST, MANAGER	Room type revenue
GET	/api/hotel-branches/{branchId}/analytics/hotel/customers	Assigned RECEPTIONIST, MANAGER	Customer analytics
GET	/api/hotel-branches/{branchId}/analytics/hotel/housekeeping	Assigned RECEPTIONIST, MANAGER	Housekeeping analytics
GET	/api/hotel-branches/{branchId}/analytics/hotel/attendance	Assigned RECEPTIONIST, MANAGER	Attendance analytics
GET	/api/hotel-branches/{branchId}/analytics/restaurant/dashboard	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Restaurant dashboard
GET	/api/hotel-branches/{branchId}/analytics/restaurant/sales	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Restaurant sales
GET	/api/hotel-branches/{branchId}/analytics/restaurant/dish-sales	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Dish sales
GET	/api/hotel-branches/{branchId}/analytics/restaurant/chef-performance	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Chef performance
GET	/api/hotel-branches/{branchId}/analytics/restaurant/serving-staff-performance	Assigned RECEPTIONIST, RESTAURANT_ADMIN, MANAGER	Serving staff performance
GET	/api/hotel-branches/{branchId}/analytics/overall/revenue	Assigned RECEPTIONIST, MANAGER	Branch overall revenue
GET	/api/manager/analytics/branches/dashboard	MANAGER	All branches dashboard
GET	/api/manager/analytics/branches/revenue	MANAGER	Branch revenue comparison
GET	/api/manager/analytics/branches/occupancy	MANAGER	Branch occupancy comparison
GET	/api/manager/analytics/branches/attendance	MANAGER	Branch attendance report
GET	/api/root-admin/analytics/website	ROOT_ADMIN	Entire website analytics
GET	/api/root-admin/analytics/revenue	ROOT_ADMIN	Entire platform revenue

22. Important API Request / Response Samples

22.1 Root Admin Creates Manager

Method & Endpoint: POST /api/root-admin/managers

Access: ROOT_ADMIN

Purpose: Create Manager under Root Admin.

Request Body / Form Data:

```
{
  "fullName": "Hotel Manager",
  "email": "manager@agacostays.com",
  "phone": "9876543210",
  "password": "Manager@123",
  "roleName": "MANAGER",
  "accessLevel": "ALL_BRANCHES"
}
```

22.2 Manager Adds Hotel Branch

Method & Endpoint: POST /api/manager/hotel-branches

Access: MANAGER

Purpose: Add AGA CoStays branch in a city.

Request Body / Form Data:

```
{
  "cityId": 1,
  "branchName": "AGA CoStays Bengaluru",
  "address": "MG Road, Bengaluru, Karnataka",
  "landmark": "Near Metro Station",
  "latitude": 12.9716,
  "longitude": 77.5946,
  "phone": "9876543210",
  "email": "bengaluru@agacostays.com",
  "description": "Premium AGA CoStays hotel located in Bengaluru city center",
  "status": "ACTIVE"
}
```

22.3 Manager Updates Room Price

Method & Endpoint: PUT /api/manager/hotel-branches/{branchId}/rooms/{roomId}/price

Access: MANAGER, ROOT_ADMIN

Purpose: Update current room price based on location.

Request Body / Form Data:

```
{
  "newPricePerDay": 1800,
  "effectiveFrom": "2026-07-10",
  "reason": "Updated price based on Bengaluru location demand"
}
```

Sample Response:

```
{
  "status": "success",
  "message": "Room price updated successfully",
  "branchId": 1,
  "roomId": 2,
  "oldPricePerDay": 1600,
  "newPricePerDay": 1800,
  "effectiveFrom": "2026-07-10"
}
```

22.4 Customer Views Rooms With Current Price

Method & Endpoint: GET /api/hotel-branches/{branchId}/rooms

Access: Public / CUSTOMER

Purpose: Customer sees room photos and location-specific current price.

Sample Response:

```
[
  {
    "roomId": 2,
    "branchId": 1,
    "branchName": "AGA CoStays Bengaluru",
```

```

    "roomNumber": "B201",
    "roomType": "DELUXE",
    "basePricePerDay": 1600,
    "currentPricePerDay": 1800,
    "description": "Semi luxury room with bed, washroom, kitchen, AC, TV and fridge",
    "status": "AVAILABLE",
    "primaryPhotoUrl": "https://agacostays.com/uploads/rooms/b201/photo1.jpg"
  }
]

```

22.5 Customer Books Room

Method & Endpoint: POST /api/hotel-branches/{branchId}/bookings

Access: CUSTOMER

Purpose: Customer books selected branch room after login/register.

Request Body / Form Data:

```

{
  "roomId": 2,
  "checkInDate": "2026-07-15",
  "checkOutDate": "2026-07-18",
  "checkInTime": "12:00",
  "checkOutTime": "11:00",
  "numberOfGuests": 2
}

```

Sample Response:

```

{
  "status": "success",
  "bookingId": 501,
  "branchId": 1,
  "branchName": "AGA CoStays Bengaluru",
  "roomNumber": "B201",
  "roomType": "DELUXE",
  "pricePerDay": 1800,
  "numberOfDays": 3,
  "totalAmount": 5400,
  "bookingStatus": "PENDING",
  "paymentStatus": "PENDING",
  "emailStatus": "BOOKING_EMAIL_SENT"
}

```

22.6 Receptionist Uploads Room Photos

Method & Endpoint: POST /api/hotel-branches/{branchId}/rooms/{roomId}/photos

Access: Assigned RECEPTIONIST, MANAGER

Purpose: Upload room photos for assigned branch.

Request Body / Form Data:

```

Content-Type: multipart/form-data
photos: file[]
caption: Deluxe room front view
isPrimary: true

```

22.7 Restaurant Admin Uploads Dish Photos

Method & Endpoint: POST /api/hotel-branches/{branchId}/restaurant/menu/{menuItemId}/photos

Access: Assigned RESTAURANT_ADMIN, MANAGER

Purpose: Upload dish photos for customers.

Request Body / Form Data:

```
Content-Type: multipart/form-data
photos: file[]
caption: Paneer Butter Masala dish photo
isPrimary: true
```

22.8 Customer Places Food Order

Method & Endpoint: POST /api/hotel-branches/{branchId}/restaurant/orders

Access: CUSTOMER

Purpose: Place restaurant food order from room.

Request Body / Form Data:

```
{
  "bookingId": 501,
  "roomId": 2,
  "items": [
    { "menuItemId": 1, "quantity": 2 },
    { "menuItemId": 2, "quantity": 1 }
  ],
  "deliveryType": "ROOM_DELIVERY",
  "paymentMode": "ONLINE"
}
```

22.9 Manager Creates Salary Structure

Method & Endpoint: POST /api/manager/payroll/salary-structures

Access: MANAGER

Purpose: Create salary structure based on role.

Request Body / Form Data:

```
{
  "roleName": "CHEF",
  "basicSalary": 25000,
  "hra": 5000,
  "allowances": 3000,
  "deductions": 1000,
  "currency": "INR",
  "effectiveFrom": "2026-07-01"
}
```

23. UI Screens by Role

This section lists the recommended screens for the application. Some screens can be reused with role-based permission controls.

23.1 Public Website Screens

No.	Screen	Access	Purpose
1	Home Page	Public	Shows banners, brand introduction, city search, featured hotels and booking CTA.
2	Cities / Locations Page	Public	Shows metropolitan cities where AGA CoStays is available.
3	Hotel Branch Listing Page	Public	Shows hotels based on selected city.
4	Hotel Branch Details Page	Public	Shows branch details, address, map, hotel photos, receptionist contacts.
5	Hotel Photo Gallery	Public	Shows hotel branch photos.

6	Room Listing Page	Public	Shows rooms, current location price, room photo and description.
7	Room Details Page	Public	Shows full room details, photos, facilities, availability and Book Now button.
8	Room Photo Gallery	Public	Shows all photos of selected room.
9	Restaurant Menu Preview	Public	Shows dishes and dish photos for selected branch.
10	Dish Details Page	Public	Shows dish description, price, photo and availability.
11	Receptionist / Emergency Contacts	Public	Shows branch support and emergency details.
12	Login / Register Page	Public	Shown when user tries protected action like Book Now.

23.2 Customer Screens

No.	Screen	Access	Purpose
1	Customer Dashboard	CUSTOMER	Summary of bookings, orders, payments and notifications.
2	Customer Profile	CUSTOMER	View and update profile.
3	Change Password	CUSTOMER	Change account password.
4	Booking Form	CUSTOMER	Enter dates, guests and confirm booking.
5	Booking Confirmation	CUSTOMER	Shows booking summary before/after booking.
6	My Bookings	CUSTOMER	List of customer bookings.
7	Booking Details	CUSTOMER	Full booking details and status.
8	Payment Page	CUSTOMER	Pay for room, food or final bill.
9	My Payments	CUSTOMER	Customer payment history.
10	Final Bill	CUSTOMER	View room + restaurant final bill.
11	Invoice	CUSTOMER	Download/view invoice.
12	Restaurant Menu	CUSTOMER	View full branch restaurant menu.
13	Cart	CUSTOMER	Selected food items before placing order.
14	Food Order Confirmation	CUSTOMER	Confirm food order.
15	My Food Orders	CUSTOMER	List of customer food orders.
16	Food Order Tracking	CUSTOMER	Track order status from placed to delivered.
17	Support Request	CUSTOMER	Raise emergency/room/food/payment support request.
18	My Support Requests	CUSTOMER	View support request status.
19	Feedback	CUSTOMER	Submit hotel and restaurant feedback.
20	Notifications	CUSTOMER	Booking, payment, checkout reminder and food notifications.

23.3 Root Admin Screens

No.	Screen	Access	Purpose
1	Root Admin Login	ROOT_ADMIN	Root Admin login screen.
2	Root Dashboard	ROOT_ADMIN	Platform-wide overview.
3	Manager Management	ROOT_ADMIN	Create/update/deactivate managers.
4	Role Management	ROOT_ADMIN	Manage roles.
5	Permission Management	ROOT_ADMIN	Manage role permissions.
6	Website Settings	ROOT_ADMIN	Website name, booking status, maintenance mode.
7	Banner Management	ROOT_ADMIN	Manage homepage banners.
8	Website Pages Management	ROOT_ADMIN	About Us, Terms, Privacy, FAQ.
9	Payment Gateway Settings	ROOT_ADMIN	Configure payment gateway.
10	Email/SMS Settings	ROOT_ADMIN	Configure notification providers.
11	All Branches Overview	ROOT_ADMIN	View all branches.
12	Complete Branch Details	ROOT_ADMIN	Full branch data overview.
13	Website Analytics	ROOT_ADMIN	Platform analytics.
14	City/Branch Performance	ROOT_ADMIN	City and branch comparison.
15	Audit Logs	ROOT_ADMIN	Track system changes.
16	Data Export	ROOT_ADMIN	Export customers, bookings, payroll, revenue.
17	Backups	ROOT_ADMIN	Create/download backups.

23.4 Manager Screens

No.	Screen	Access	Purpose
1	Manager Login	MANAGER	Manager login.
2	Manager Dashboard	MANAGER	Business dashboard.
3	City Management	MANAGER	Add/update/deactivate cities.
4	Hotel Branch Management	MANAGER	Manage hotel branches.
5	Branch Photo Management	MANAGER	Upload/update/delete branch photos.
6	Staff Management	MANAGER	Create/update/deactivate staff.
7	Staff Branch Assignment	MANAGER	Assign or transfer staff to branch.
8	Customer Management	MANAGER	View and manage customer status.
9	Room Management	MANAGER	Manage rooms.

10	Room Photo Management	MANAGER	Manage room photos.
11	Room Description/Status	MANAGER	Update room description/status.
12	Location-Based Room Pricing	MANAGER	Update room price by branch.
13	Room Price History	MANAGER	View price change history.
14	Booking Management	MANAGER	View/approve/cancel/check-in/check-out bookings.
15	Billing and Payments	MANAGER	View bills, payments and refunds.
16	Restaurant Overview	MANAGER	View branch restaurant details.
17	Restaurant Menu	MANAGER	Manage dishes.
18	Dish Photos	MANAGER	Manage dish photos.
19	Restaurant Orders	MANAGER	Manage food orders.
20	Assign Chef/Serving Staff	MANAGER	Assign restaurant staff.
21	Attendance	MANAGER	Working/absent/monthly attendance.
22	Salary Structure	MANAGER	Create role-based salary.
23	Staff Bank Accounts	MANAGER	Manage salary bank details.
24	Generate Payroll	MANAGER	Generate monthly payroll.
25	Credit Salaries	MANAGER	Credit salary at month end.
26	Salary Slips	MANAGER	Generate/download/send salary slips.
27	Payroll Reports	MANAGER	Payroll reports.
28	Hotel Analytics	MANAGER	Hotel analytics.
29	Restaurant Analytics	MANAGER	Restaurant analytics.
30	Overall Revenue Analytics	MANAGER	Room + restaurant revenue.

23.5 Receptionist Screens

No.	Screen	Access	Purpose
1	Receptionist Login	RECEPTIONIST	Login.
2	Receptionist Dashboard	RECEPTIONIST	Assigned branch summary.
3	My Branch Details	RECEPTIONIST	View assigned branch.
4	Branch Rooms	RECEPTIONIST	View/add/update rooms except price.
5	Room Photos Upload	RECEPTIONIST	Upload room photos.
6	Room Description Update	RECEPTIONIST	Update room description.
7	Room Status Update	RECEPTIONIST	Update room status.
8	Booking Requests	RECEPTIONIST	Approve/reject bookings.
9	Booking Details	RECEPTIONIST	View booking.
10	Customer Check-In	RECEPTIONIST	Check-in customer.
11	Customer Checkout	RECEPTIONIST	Checkout customer.
12	Customer List	RECEPTIONIST	View customers.
13	Support Requests	RECEPTIONIST	Handle customer support.
14	Restaurant Orders	RECEPTIONIST	View/accept/reject food orders.
15	Place Food Order for Customer	RECEPTIONIST	Assist customer order.
16	Assign Chef	RECEPTIONIST	Assign chef.
17	Assign Serving Staff	RECEPTIONIST	Assign delivery staff.
18	Working Staff Today	RECEPTIONIST	View present staff.
19	Absent Staff Today	RECEPTIONIST	View absent staff.
20	Monthly Attendance	RECEPTIONIST	View attendance.
21	Branch Analytics	RECEPTIONIST	Hotel/restaurant/revenue analytics.

23.6 Restaurant Admin Screens

No.	Screen	Access	Purpose
1	Restaurant Admin Login	RESTAURANT_ADMIN	Login.
2	Restaurant Admin Dashboard	RESTAURANT_ADMIN	Restaurant summary.
3	My Branch Restaurant	RESTAURANT_ADMIN	View restaurant details.
4	Menu Management	RESTAURANT_ADMIN	Manage dishes.
5	Add Dish	RESTAURANT_ADMIN	Add menu item.
6	Update Dish	RESTAURANT_ADMIN	Update dish info.
7	Dish Photo Management	RESTAURANT_ADMIN	Upload/update/delete dish photos.
8	Dish Availability	RESTAURANT_ADMIN	Set dish availability.
9	Restaurant Orders	RESTAURANT_ADMIN	Manage orders.
10	Order Details	RESTAURANT_ADMIN	View order.
11	Accept/Reject Order	RESTAURANT_ADMIN	Accept/reject order.
12	Assign Chef	RESTAURANT_ADMIN	Assign chef.
13	Assign Serving Staff	RESTAURANT_ADMIN	Assign serving staff.
14	Restaurant Feedback	RESTAURANT_ADMIN	View feedback.
15	Restaurant Sales Analytics	RESTAURANT_ADMIN	View sales.
16	Dish-Wise Sales	RESTAURANT_ADMIN	Sales by dish.
17	Chef Performance	RESTAURANT_ADMIN	Chef report.
18	Serving Staff Performance	RESTAURANT_ADMIN	Delivery report.

23.7 Chef Screens

No.	Screen	Access	Purpose
1	Chef Login	CHEF	Login.
2	Chef Dashboard	CHEF	Kitchen summary.
3	My Kitchen Orders	CHEF	View assigned orders.
4	Order Details	CHEF	View items.

5	Start Preparing	CHEF	Update order to preparing.
6	Mark Food Ready	CHEF	Update ready for delivery.
7	Kitchen Notes	CHEF	Add preparation notes.
8	Attendance Check-In/Out	CHEF	Mark attendance.
9	My Attendance	CHEF	View own attendance.

23.8 Serving Staff Screens

No.	Screen	Access	Purpose
1	Serving Staff Login	SERVING_STAFF	Login.
2	Serving Staff Dashboard	SERVING_STAFF	Delivery summary.
3	My Delivery Orders	SERVING_STAFF	View assigned deliveries.
4	Order Details	SERVING_STAFF	View order details.
5	Pick Up Food	SERVING_STAFF	Pickup from kitchen.
6	Mark Delivered	SERVING_STAFF	Mark delivered to room.
7	Report Delivery Issue	SERVING_STAFF	Report issue.
8	Attendance Check-In/Out	SERVING_STAFF	Mark attendance.
9	My Attendance	SERVING_STAFF	View own attendance.

23.9 Housekeeping Screens

No.	Screen	Access	Purpose
1	Housekeeping Login	HOUSEKEEPING_STAFF	Login.
2	Housekeeping Dashboard	HOUSEKEEPING_STAFF	Task summary.
3	My Cleaning Tasks	HOUSEKEEPING_STAFF	View tasks.
4	Cleaning Task Details	HOUSEKEEPING_STAFF	View task details.
5	Update Cleaning Status	HOUSEKEEPING_STAFF	Update completed/pending.
6	Attendance Check-In/Out	HOUSEKEEPING_STAFF	Mark attendance.
7	My Attendance	HOUSEKEEPING_STAFF	View own attendance.

23.10 Common Screens

No.	Screen	Access	Purpose
1	Login	All roles	Common login screen.
2	Register	Customer	Customer registration.
3	Forgot Password	All roles	Password recovery.
4	Reset Password	All roles	Reset password.
5	Change Password	All roles	Change password.
6	Profile	All logged-in users	View/update profile based on role.
7	Notifications	Logged-in users	View notifications.
8	Unauthorized	All roles	Access denied page.
9	Not Found / Error	All roles	Routing or server error page.

24. UI Flow Diagrams

24.1 Public Customer Browsing Flow

Home

- > Select City
 - > Hotel Branch Listing
 - > Hotel Branch Details
 - > Hotel Photos
 - > Receptionist/Emergency Contact
 - > Room Listing
 - > Room Details
 - > Room Photos
 - > Book Now
 - > If Not Logged In: Login/Register
 - > If Logged In: Booking Form

24.2 Customer Booking Flow

Room Details

- > Book Now
 - > Login/Register if not authenticated
 - > Return to Booking Form
 - > Select Check-in/Checkout Dates
 - > Confirm Booking
 - > Payment Page
 - > Booking Confirmation

- > Booking Email Sent
- > My Bookings

24.3 Customer Restaurant Order Flow

Hotel Branch Details

- > Restaurant Menu
 - > Dish Details / Dish Photos
 - > Add to Cart
 - > Login/Register if not authenticated
 - > Place Food Order
 - > Payment or Add to Final Bill
 - > Order Tracking
 - > Delivered
 - > Restaurant Feedback

24.4 Receptionist Branch Flow

Receptionist Login

- > Receptionist Dashboard
 - > Booking Requests
 - > Approve/Reject
 - > Check-In
 - > Checkout
 - > Rooms
 - > Add/Update Room
 - > Upload Room Photos
 - > Update Description/Status
- > Restaurant Orders
 - > Accept/Reject
 - > Assign Chef
 - > Assign Serving Staff
- > Attendance
 - > Working Staff
 - > Absent Staff
 - > Monthly Attendance
- > Branch Analytics

24.5 Restaurant Admin Flow

Restaurant Admin Login

- > Restaurant Dashboard
 - > Menu Management
 - > Add/Update Dish
 - > Upload/Update/Delete Dish Photos
 - > Update Availability
- > Restaurant Orders
 - > Accept/Reject Order
 - > Assign Chef
 - > Assign Serving Staff
- > Feedback
- > Restaurant Analytics

24.6 Chef Flow

Chef Login

- > Chef Dashboard
 - > My Kitchen Orders
 - > Order Details
 - > Start Preparing
 - > Add Kitchen Notes
 - > Mark Food Ready
- > Attendance Check-In/Out

24.7 Serving Staff Flow

Serving Staff Login

- > Delivery Dashboard
 - > My Delivery Orders

- > Order Details
- > Pick Up Food
 - > Deliver to Room
 - > Mark Delivered / Report Issue
- > Attendance Check-In/Out

24.8 Housekeeping Flow

Housekeeping Login

- > Housekeeping Dashboard
- > My Cleaning Tasks
 - > Task Details
 - > Clean Room and Washroom
 - > Update Status Completed
- > Attendance Check-In/Out

24.9 Manager Flow

Manager Login

- > Manager Dashboard
- > City / Branch Management
- > Staff Management and Branch Assignment
- > Room Management
 - > Room Photos
 - > Room Description
 - > Location-Based Price Management
 - > Price History
- > Booking Management
- > Restaurant Management
- > Attendance
- > Payroll
 - > Salary Structure
 - > Bank Account
 - > Generate Payroll
 - > Credit Salaries
 - > Salary Slips
- > Reports and Analytics

24.10 Root Admin Flow

Root Admin Login

- > Root Dashboard
- > Manager Management
- > Roles and Permissions
- > Website Settings
- > Website Banners and Pages
- > Payment Gateway Settings
- > Email/SMS Settings
- > All Branches Overview
- > Platform Analytics
- > Audit Logs
- > Data Export
- > Backups

25. Screen Count Summary

Module / Role	Approx. Screens	Notes
Public Website	12	Accessible without login
Customer	20	Protected screens after login
Root Admin	17	Platform-level screens
Manager	30	Business operations screens
Receptionist	21	Assigned branch screens
Restaurant Admin	18	Assigned restaurant screens
Chef	9	Kitchen screens
Serving Staff	9	Delivery screens
Housekeeping Staff	7	Cleaning screens
Common Screens	9	Reusable screens
Total Unique/Planned Screens	Approx. 100-125	Some screens are reusable by role

26. Database Entities / Tables

roles

Allowed roles: ROOT_ADMIN, MANAGER, CUSTOMER, RECEPTIONIST, RESTAURANT_ADMIN, CHEF, SERVING_STAFF, HOUSEKEEPING_STAFF.

```
role_id
role_name
description
created_at
updated_at
```

users

```
user_id
full_name
email
phone
password_hash
role_id
user_type
status
created_at
updated_at
```

root_admin_profiles

```
root_admin_id
user_id
full_name
email
phone
status
created_at
updated_at
```

manager_profiles

Replaces old admin_profiles.

```
manager_id
user_id
full_name
email
phone
access_level
status
created_by_root_admin
created_at
updated_at
```

permissions

```
permission_id
permission_code
permission_name
description
module_name
created_at
updated_at
```

role_permissions

```
role_permission_id
role_id
permission_id
```

created_by
created_at

cities

city_id
city_name
state
country
status
created_at
updated_at

hotel_branches

branch_id
city_id
branch_name
address
landmark
latitude
longitude
phone
email
description
status
created_by
created_at
updated_at

hotel_branch_photos

photo_id
branch_id
photo_url
caption
is_primary
uploaded_by
created_at
updated_at

staff

staff_id
user_id
department
shift
joining_date
status
created_at
updated_at

staff_branch_mapping

Controls branch-level access for staff.

mapping_id
staff_id
branch_id
role_id
department
shift
assigned_from
assigned_to
status
created_by
created_at
updated_at

customers

customer_id
user_id
address
status
created_at
updated_at

rooms

Use current_price_per_day for customer display and booking calculation.

room_id
branch_id
room_number
room_type
base_price_per_day
current_price_per_day
description
floor
status
created_by
updated_by
created_at
updated_at

room_price_history

Tracks every room price change.

price_history_id
room_id
branch_id
old_price
new_price
changed_by
changed_by_role
change_reason
effective_from
created_at

room_photos

photo_id
branch_id
room_id
photo_url
caption
is_primary
uploaded_by
created_at
updated_at

bookings

price_per_day_at_booking freezes the booking price.

booking_id
branch_id
customer_id
room_id
check_in_date
check_out_date
check_in_time
check_out_time
number_of_guests
number_of_days

price_per_day_at_booking
room_charges
booking_status
payment_status
created_at
updated_at

restaurants

restaurant_id
branch_id
restaurant_name
description
opening_time
closing_time
status
created_at
updated_at

restaurant_menu

menu_item_id
branch_id
restaurant_id
item_name
category
description
price
availability
food_type
preparation_time_minutes
created_by
created_at
updated_at

restaurant_menu_photos

photo_id
branch_id
restaurant_id
menu_item_id
photo_url
caption
is_primary
uploaded_by
created_at
updated_at

restaurant_orders

order_id
branch_id
restaurant_id
booking_id
customer_id
room_id
room_number
order_amount
delivery_type
order_status
payment_status
assigned_chef_id
assigned_serving_staff_id
ordered_at
updated_at

restaurant_order_items

order_item_id
order_id
menu_item_id
quantity
price
total_amount

payments

payment_id
branch_id
payment_for
reference_id
customer_id
amount
currency
payment_method
payment_status
gateway_order_id
gateway_payment_id
gateway_signature
paid_at
created_at

billing

bill_id
branch_id
booking_id
customer_id
room_charges
restaurant_charges
tax_amount
discount
final_amount
payment_status
invoice_url
created_at

feedback

feedback_id
branch_id
booking_id
customer_id
hotel_rating
hotel_comments
restaurant_rating
restaurant_comments
created_at

restaurant_feedback

feedback_id
branch_id
booking_id
order_id
customer_id
food_rating
taste_rating
delivery_rating
comments
created_at

housekeeping_tasks

task_id
branch_id
room_id
task_type
description
assigned_to
cleaning_date
status
created_by
created_at
updated_at

attendance

attendance_id
branch_id
staff_id
attendance_date
status
check_in_time
check_out_time
worked_hours
remarks
marked_by
created_at
updated_at

receptionist_contacts

contact_id
branch_id
staff_id
phone
alternate_phone
email
shift
available_from
available_to
purpose
is_emergency_contact
status
created_at
updated_at

customer_support_requests

request_id
branch_id
customer_id
booking_id
request_type
subject
description
priority
status
assigned_to
created_at
updated_at

notification_logs

notification_id
branch_id
customer_id
booking_id
notification_type
title

message
email_to
email_status
is_read
sent_at
created_at

checkout_reminders

reminder_id
branch_id
booking_id
customer_id
checkout_date
checkout_time
reminder_before_hours
scheduled_time
sent_time
status
created_at
updated_at

salary_structures

salary_structure_id
role_id
role_name
basic_salary
hra
allowances
deductions
monthly_gross_salary
monthly_net_salary
currency
effective_from
status
created_by
created_at
updated_at

staff_bank_accounts

bank_account_id
staff_id
account_holder_name
bank_name
account_number
ifsc_code
branch_name
account_type
status
created_by
created_at
updated_at

payroll

payroll_id
branch_id
staff_id
role_id
department
month
year
total_working_days
present_days
absent_days
half_days
leave_days

basic_salary
hra
allowances
bonus
gross_salary
attendance_deduction
other_deductions
net_salary
payment_status
payment_mode
transaction_id
paid_at
remarks
generated_by
created_at
updated_at

payroll_payment_batches

payment_batch_id
branch_id
month
year
total_staff
total_amount
payment_mode
status
created_by
created_at
completed_at

salary_slips

salary_slip_id
payroll_id
staff_id
month
year
salary_slip_url
generated_at
sent_email_status
created_at

website_settings

setting_id
website_name
support_email
support_phone
default_currency
booking_enabled
restaurant_ordering_enabled
maintenance_mode
maintenance_message
updated_by
updated_at

website_banners

banner_id
title
subtitle
image_url
redirect_url
display_order
status
created_by
created_at
updated_at

website_pages

page_type values: ABOUT_US, TERMS_AND_CONDITIONS, PRIVACY_POLICY, CONTACT_US, FAQ.

```
page_id
page_type
title
content
status
updated_by
created_at
updated_at
```

payment_gateway_settings

Secrets must be encrypted and never returned in plain text.

```
gateway_setting_id
gateway_name
key_id
encrypted_key_secret
encrypted_webhook_secret
status
created_by
updated_by
created_at
updated_at
```

email_settings

```
email_setting_id
provider
host
port
username
encrypted_password
from_email
status
created_by
updated_by
created_at
updated_at
```

sms_settings

```
sms_setting_id
provider
api_key_encrypted
sender_id
status
created_by
updated_by
created_at
updated_at
```

audit_logs

```
audit_log_id
user_id
role_name
module_name
action
description
old_value
new_value
ip_address
user_agent
created_at
```

system_backups

```
backup_id
backup_type
backup_file_url
backup_status
remarks
created_by
created_at
completed_at
```

27. Entity Relationship Summary

- users.role_id references roles.role_id.
- root_admin_profiles.user_id references users.user_id.
- manager_profiles.user_id references users.user_id.
- staff.user_id references users.user_id.
- customers.user_id references users.user_id.
- hotel_branches.city_id references cities.city_id.
- hotel_branch_photos.branch_id references hotel_branches.branch_id.
- staff_branch_mapping maps staff to one or more branches and controls branch access.
- rooms.branch_id references hotel_branches.branch_id.
- room_price_history references room_id and branch_id.
- bookings reference branch_id, customer_id and room_id.
- bookings.price_per_day_at_booking stores the exact price used during booking.
- restaurants.branch_id references hotel_branches.branch_id.
- restaurant_menu and restaurant_menu_photos are branch-specific.
- restaurant_orders are linked to branch, restaurant, booking, room and customer.
- payments use payment_for and reference_id to connect to room booking, food order or final bill.
- attendance and payroll are branch-specific.
- notification_logs and checkout_reminders reference branch and booking details.

28. Middleware Authorization Logic

```
if role == ROOT_ADMIN:
    allow all actions across all branches and website root modules

elif role == MANAGER:
    allow business modules for assigned/all branches based on access_level
    deny root-level secrets/settings unless explicit permission is granted

elif role in [RECEPTIONIST, RESTAURANT_ADMIN, CHEF, SERVING_STAFF, HOUSEKEEPING_STAFF]:
    allow only if requested branchId == loggedInUser.assignedBranchId

elif role == CUSTOMER:
    allow public branch/hotel/room/restaurant browsing
    allow booking/order/payment/bill/feedback only if loggedInCustomerId owns the resource

else:
    deny access
```

29. Booking Email Required Fields

- Customer Name
- Booking ID

- Hotel Branch Name
- City
- Hotel Address
- Latitude/Longitude or Map Link
- Receptionist Contact Number
- Emergency Contact Number
- Room Number
- Room Type
- Room Photos
- Room Description
- Price Per Day Used For Booking
- Check-in Date and Time
- Checkout Date and Time
- Payment Status
- Payment Link if Pending
- Total Booking Amount

30. API Prefix Summary

Root Admin:

/api/root-admin/...

Manager:

/api/manager/...

Public / Customer browsing:

/api/cities

/api/hotel-branches

/api/hotel-branches/{branchId}/...

Branch Rooms:

/api/hotel-branches/{branchId}/rooms

/api/hotel-branches/{branchId}/rooms/{roomId}/photos

Manager Room Pricing:

/api/manager/hotel-branches/{branchId}/rooms/{roomId}/price

/api/manager/hotel-branches/{branchId}/room-types/{roomType}/price

/api/manager/hotel-branches/{branchId}/rooms/{roomId}/price-history

Branch Restaurant:

/api/hotel-branches/{branchId}/restaurant

/api/hotel-branches/{branchId}/restaurant/menu

/api/hotel-branches/{branchId}/restaurant/orders

Staff Own Branch:

/api/staff/my-branch/...

Customer Protected:

/api/customers/...

/api/bookings/my-bookings

/api/restaurant/orders/my-orders

/api/payments/my-payments

Payroll:

/api/manager/payroll/...

/api/manager/hotel-branches/{branchId}/payroll/...

Analytics:

/api/root-admin/analytics/...

```
/api/manager/analytics/...  
/api/hotel-branches/{branchId}/analytics/...
```

31. Recommended Development Phases

- Phase 1: Public website, customer browsing, cities, branches, rooms, photos, login/register and booking.
- Phase 2: Manager dashboard, branch management, staff management, room price management and receptionist booking operations.
- Phase 3: Restaurant menu, dish photos, customer food ordering, chef and serving staff workflow.
- Phase 4: Payments, billing, invoices, email notifications, checkout reminders and feedback.
- Phase 5: Attendance, payroll, salary credit and salary slips.
- Phase 6: Root Admin website settings, analytics, exports, backups and audit logs.